

DRAFT: The Scalar-Angular-Torsion Framework: I. Ontological Foundation of the Zottenwelt

Nathan McKnight
Independent Researcher
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We introduce the Scalar-Angular-Torsion (SAT) framework, a zero-parameter geometric ontology that derives 3D physical laws from the mechanics of 4D filamental worldlines. By re-architecting the foundational substrate of physics as a hypercomplex fabric of physical filaments—the Zottenwelt—we move from the “geometric expense” of probabilistic field theories toward a rigid, deterministic 4D geometry. We define the four foundational pillars of this ontology: Filamental Primacy, Intersectional Trace, Zero-Parameter Economy, and Failure Rigidity. This paper establishes the coordinate engine and the Universal Indicatrix (UI) mapping required to project 4D filamental dynamics into observable 3D manifolds.

LAY OVERVIEW

The Scalar-Angular-Torsion (SAT) framework begins with a fundamental change in how we view reality, moving away from the abstract “fields” of modern physics and toward a physical, geometric world. Imagine the universe not as an empty void, but as a dense, tangled fabric of four-dimensional “filaments.” We call this substrate the **Zottenwelt**. In this view, what we perceive as 3D particles are actually just the points where these 4D filaments pierce through our current moment in time—much like a needle passing through a sheet of paper.

This approach is governed by a “**Zero-Parameter Economy**.” In current physics, scientists manually “tune” dozens of numbers (like the mass of an electron) to make their equations work. In the SAT framework, these numbers aren’t chosen; they are mandatory results of the 4D geometry. If the geometry of the filaments and the universal lattice is correct, all physical constants “snap” into place automatically. This creates a “**Failure Rigidity**”: the theory is designed to be either entirely correct or fundamentally broken, leaving no room for manual adjustments to hide errors.

I. INTRODUCTION: THE ONTOLOGICAL SHIFT

The current paradigm of theoretical physics relies on a collection of effectively independent fields and manually tuned coupling constants to describe the Standard Model. The Scalar-Angular-Torsion (SAT) framework proposes a departure from this “geometric expense” in favor of a “Zero-Parameter Economy”. This paradigm shift identifies 3D particles not as independent entities or point-like excitations, but as structural inevitabilities of a rigid 4D geometry.

At the core of this framework is the **Zottenwelt**: a hypercomplex tangled fabric of physical 4D worldline filaments. Within this substrate, the worldline is the primary physical unit; attributes conventionally treated as intrinsic, such as mass or charge, are reinterpreted as

integrated geometric invariants of the filament’s coiling history.

II. THE FOUR FOUNDATIONAL PILLARS

The structural integrity of the SAT architecture is secured by four irreducible pillars that constrain the emergence of physical laws.

A. Filamental Primacy

Worldlines are not abstract mathematical trajectories but physical 4D objects possessing intrinsic tension T . These filaments supersede the probabilistic field descriptions of quantum mechanics, serving as the primary ontological substrate.

B. Intersectional Trace

Perceived 3D particles are the statistical intersections where 4D filaments pierce a propagating 3-sphere “time surface” Σ_t . Matter is thus a 3D slice of a 4D filamental ensemble.

C. Zero-Parameter Economy

Physical constants are not manually tuned but emerge as geometric invariants of a 24-cell Hypersphere Unit Cell Vertex (HSUCV) lattice. All physical units are derived from a single anchor: the Filament Scale ($l_f \approx 0.7937$ fm) pinned to the Rydberg Constant.

D. Failure Rigidity

The framework is structurally locked. Any modification to fundamental anchors, such as the Filament Scale $l_f \approx 0.7937$ fm or the Projection Constant $B \approx 0.2387$

rad, ripples catastrophically through the derived constants. This ensures the theory is either entirely correct or fundamentally broken.

III. THE COORDINATE ENGINE AND UNIVERSAL INDICATRIX

The universe is modeled as a discretized 4D hypersphere expanding at a radial rate c , which we identify as the “speed of time”. The coordinate grid is provided by the 24-cell HSUCV lattice, which serves as the universal frame of reference.

The mapping of 4D filamental trajectories into macroscopic 3D observables is governed by the **Universal Indicatrix (UI)**. The UI function $y^\mu(\lambda)$ translates the 4D hyperhelical path x_0^ν into a 3D projection through a combination of scaling $r(\lambda)$ and rotation $R_\nu^\mu(\lambda)$:

$$y^\mu(\lambda) = r(\lambda)R_\nu^\mu(\lambda)x_0^\nu \quad (1)$$

where λ represents the parametrization along the superhelical filament. The radial expansion is set to linear time expansion $r(\lambda) = \lambda = ct$.

The dynamics of this projection are governed by the

UI Lagrangian L_{UI} :

$$L_{UI} = \frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{2} r^2 \Omega_{\mu\nu} \Omega^{\mu\nu} \quad (2)$$

where $\Omega_{\mu\nu}$ is the filament angular velocity tensor. In the ground state (identity rotation and linear expansion), the Lagrangian resolves to a constant geometric effort:

$$L_{UI}^{\text{vacuum}} = 0.5 \text{ units} \quad (3)$$

This baseline ensures that the projection remains UV-finite and free of mathematical singularities.

IV. CONCLUSION OF PART I

By redefining the vacuum as a structured filamental lattice and particles as intersectional traces, the SAT framework provides a rigid geometric substrate for the derivation of physical laws. Part II will detail the discretization of this substrate through the 24-cell HSUCV lattice and the definition of the universal scale anchors.